# Multimodal Visual-Language Prompt Network for Remote Sensing Few-Shot Segmentation

Zhenhao Yang, Fukun Bi, Jianhong Han, Xianping Ma, *Member, IEEE*, Chenglong He, and Wenkai Liu

***Abstract*— Few-shot segmentation (FSS) aims to segment objects of interest in a query image using a limited set of support images. However, most existing FSS methods are designed for natural images. When extended to remote sensing scenes characterized by extreme intraclass variations and complex backgrounds, these methods struggle to provide robust segmentation guidance, leading to severe performance degradation. To address the aforementioned issues, we propose a multimodal visual-language prompt network (MVLPNet), which employs a collaborative optimization strategy for visual–textual features to tackle the remote sensing FSS task. Specifically, MVLPNet consists of a textual–visual consistency enhancement (TVCE) module and a prototype-guided semantic alignment (PGSA) module. To overcome the limited support set for better guiding the query segmentation, we propose a TVCE module that leverages the contrastive language-image pretraining (CLIP) model to capture category-specific text embeddings. An optimal transport (OT) plan is then established to tightly align these text embeddings with the visual features of the query image, thereby extracting semantic information from the query image itself to mitigate the extreme intraclass variation in remote sensing images. Furthermore, a PGSA module is proposed to suppress interference caused by complex background regions. By aggregating lost foreground regions, more comprehensive support features are extracted. Then, the query and support features are precisely matched to activate consistent foreground regions, rather than ambiguously matching the query features via a single prototype or multiple prototypes. Extensive experiments on the iSAID-5i and LoveDA-2i datasets have demonstrated that our method achieves the state of the art. The code is available at https://github.com/Gritiii/MVLPNet**

***Index Terms*— Few-shot segmentation (FSS), remote sensing, semantic segmentation, text prompts.**

## I. Introduction

SEMANTIC segmentation of remote sensing images is essential for enhancing our understanding of geographical and ecological environments, making it a critical task for applications such as mapping and observation [1], [2]. In recent years, the advent of deep learning techniques, particularly convolutional neural networks (CNNs), has significantly increased accuracy. However, the success of these methods [3], [4], [5], [6], [7], [8], [9], [10], [11] largely depends on large-scale annotated datasets, which are time-consuming and labor-intensive to create, especially for remote sensing images that are difficult to interpret. Although semi-supervised [12], [13], [14] and weakly supervised [15], [16], [17] approaches can partially alleviate the reliance on labeled data, they often struggle to generalize effectively to unseen domains due to limitations in their training paradigms.

Against this backdrop, few-shot segmentation (FSS), which relies on a few support images to segment the corresponding objects in a query image, has emerged as a key research focus. The current mainstream FSS methods are divided into prototype-based approaches [18], [19], [20] and affinity learning-based approaches [21], [22]. The former typically requires extracting one or more support prototypes from the support image, which are then used to segment the query image through comparison or aggregation. The latter directly incorporates the similarity of support–query pixel pairs to aggregate support semantics into the query feature.

Although the aforementioned methods have achieved desirable results in natural scenes, their performance deteriorates when extended to complex remote sensing scenarios. First, extreme intraclass variations in remote sensing images cause significant differences between support and query images, as shown in Fig. 1, the support image contains only part of the plane's fuselage and tail, whereas the query image includes multiple entire planes. Existing methods [23], [24], [25], [26] typically align support features with query features based on one-way similarity, overlooking inconsistencies between the support and query foregrounds. Consequently, these methods activate only part of the target region, failing to achieve complete segmentation. Compared with visual features, textual features more effectively capture category-specific information, thereby alleviating the limitations of support samples in covering intraclass diversity. Therefore, we introduce the contrastive language-image pretraining (CLIP) model [27] to fully leverage the power of textual prompts in assisting remote sensing FSS segmentation. In addition, remote sensing images are often affected by interference from complex background regions [5], [28], as shown in Fig. 1(a), the airport area contains background elements that visually resemble planes, such as "cars" and "boarding bridges." However, existing methods [24], [25], [29], [30] have focused on ambiguously

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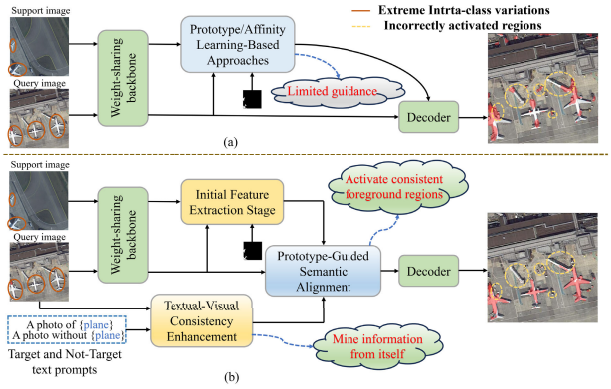



Fig. 1. Comparison between (a) existing FSS methods and (b) proposed MVLPNet in handling intraclass variations and background interference. (a) Existing methods rely on support images to guide the segmentation of query images but overlook the extreme intraclass variations in remote sensing images. Due to limited visual cues, these methods struggle to establish robust class-specific representations, resulting in incomplete segmentation. In addition, interference from background regions causes misclassification of objects in the background, such as "boarding bridges" and "cars" as foreground "planes." (b) In contrast, our proposed MVLPNet employs a collaborative optimization strategy for visual–textual features to tackle the remote sensing FSS task, effectively overcoming these challenges. MVLPNet not only extracts category semantics to enhance the target region features from the query image itself but also explicitly guides the model to focus more on the consistent foreground regions between the query and support images rather than the background ones.

matching query features using one or multiple prototypes for segmentation, overlooking interference from similar backgrounds, which leads to incorrect activations.

To address the above challenges, we propose a multimodal visual-language prompt network (MVLPNet), which employs a collaborative optimization strategy for visual–textual features to tackle the remote sensing FSS task, as shown in Fig. 1(b). Specifically, we first propose a textual–visual consistency enhancement (TVCE) module that leverages CLIP to extract category-specific text embeddings and visual features from the query image. Notably, textual features are solely used to represent category semantics and remain unaffected by variations in individual samples. Therefore, integrating textual features can effectively compensate for the category diversity that a single sample fails to capture. Then, different from existing methods [31], [32] that directly align two modalities, we model the feature alignment between text and image modalities as an optimal transport (OT) plan, which uses fine-grained visual features to update the text embeddings. In this way, the textual embeddings can be associated with the most similar visual features, achieving fine-grained semantic alignment. Finally, the alignment features are input into Softmax-GradCAM [33], compelling the model to determine whether a pixel belongs to the target class. By leveraging the powerful visual and textual understanding capabilities of CLIP, the TVCE module extracts semantic information from the query image itself, thus providing more robust guidance for addressing the extreme intraclass variations in remote sensing images.

Furthermore, we propose a prototype-guided semantic alignment (PGSA) module that mines the common foreground semantics from the support–query image pair to achieve desirable segmentation. Specifically, PGSA consists of the support information aggregation (SIA) and prototype correction (PC) modules for precise feature matching. First, the SIA module makes an initial prediction on the support features, encoding both covered and uncovered foreground regions into primary and auxiliary support vectors. By aggregating these vectors, we extract more comprehensive support features, thereby providing more precise guidance for subsequent feature matching. Next, the PC module generates support foreground and background prototypes via mask average pooling (MAP) and uses them to activate the corresponding regions in the query features, thus producing query foreground and background prior masks. Then, the difference between these two masks is computed to eliminate ambiguous query regions that exhibit high similarity to both support foreground and background features, thus further generating the most discriminative query foreground prototype. Finally, this prototype is employed to activate the query regions most similar to the support foreground while suppressing interference from background regions.

The main contributions of this article are as follows.

1) A novel MVLPNet for FSS in remote sensing is proposed to adaptively pay more attention to the query image itself based on textual prompts and to precisely match the query features with the support features, thus yielding promising results.
2) A TVCE module is proposed that tightly aligns category-specific text embeddings with the visual features of the query image based on the OT plan, thereby mining category semantics from the query image itself and providing more robust guidance for segmenting target regions.
3) We propose a PGSA module, which first aggregates lost foreground regions to extract more comprehensive support features. Subsequently, the query and support features are precisely matched to guide the model to focus more on the consistent foreground regions rather than the background ones.
4) Extensive experiments on remote sensing FSS datasets, iSAID-5i and LoveDA-2i, validate the superiority of the proposed MVLPNet, achieving the state-of-the-art performance on both datasets.

## II. Related Work

### A. Generic Semantic Segmentation

Semantic segmentation is a fundamental task in computer vision, aiming to assign an appropriate class label to each pixel in an image. Fully convolutional networks (FCNs) [34] are the first methods used to achieve effective semantic segmentation in an end-to-end manner. Subsequently, U-Net [35] introduces a symmetric encoder–decoder architecture with skip connections, allowing the decoder to retain spatial information from high-level features. On the other hand, PSPNet [36] fuses a pyramid pooling module to capture detailed object features. DeepLab [37] introduces the atrous spatial pyramid pooling (ASPP) module and incorporates dilated convolutions [38] into the FCN architecture, achieving outstanding results. However, due to the limited receptive field of CNNs, there are inherent limitations in modeling global information. To address this

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issue, several methods [4], [5], [39], [40], [41], [42] integrate a transformer into the network architecture to capture global contextual information. While transformer excels at capturing global dependencies, it also introduces significant computational complexity. Recently, Mamba [43], which is based on state-space models, has emerged as an alternative for modeling long-range dependencies. RS3Mamba [44] extends the Mamba approach to the domain of remote sensing image segmentation. Furthermore, several studies [45], [46], [47] have explored the use of multimodal techniques in the remote sensing field. In particular, Yang et al. [46] proposed a novel framework called HCNet, which efficiently utilizes multiscale visual features to achieve cross-modal alignment between image and text features, providing valuable guidance for multimodal semantic segmentation. Despite these advancements, these methods still require large amounts of annotated images and time-consuming pixel-level labels to achieve satisfactory results. Furthermore, without fine-tuning, they struggle to generalize to unseen categories.

### B. Few-Shot Learning (FSL)

Recently, FSL has emerged as a promising solution to these challenges, aiming to train a model capable of effectively recognizing new classes with limited data. Due to its cost-effectiveness, FSL has attracted significant research interest in recent years. Currently, FSL methods can be broadly categorized into three main types. The first type, transfer learning-based methods, involves extensive training on base classes followed by fine-tuning on new classes [48]. However, these approaches rely heavily on large-scale base class data, limiting their generalizability. The second type is metric learning-based methods [49], [50], [51], where the model learns a similarity measure to effectively compare and classify unseen classes using a small number of samples. The third type is metalearning-based methods [52], [53], [54], which aim to learn transferable metaknowledge from a series of tasks (i.e., episodes) sampled from the base dataset (training set), enabling the model to generalize to new tasks. Currently, most methods are based on the metalearning framework. Furthermore, due to the extreme intraclass variation in remote sensing images, several FSL methods [55], [56] specifically tailored for remote sensing tasks have been proposed. Although these methods have demonstrated outstanding performance in classification tasks, they fail to effectively address the more challenging remote sensing FSS task.

### C. Few-Shot Semantic Segmentation

FSS is a natural extension of FSL, which aims at reducing the reliance on large-scale labeled datasets. The OSLSM [57] first introduces the FSS task and establishes the foundation for subsequent research. Current FSS methods are generally categorized into two types: prototype-based methods and affinity learning-based methods. The former methods [18], [19], [58], [59] employ the MAP operation to compress support features into prototypes, which are then used to segment the query image. For example, PLNet [18] first applies prototype learning to the FSS task, using prototype embedding vectors to assist segmentation. However, relying on a single prototype for assistance often leads to the loss of local information. PMMs [19] subsequently use the expectation–maximization (EM) algorithm to generate multiple prototypes to alleviate this issue. Although these methods have achieved good results, directly compressing features into prototypes can disrupt the spatial structure of the target and result in the loss of local detail information of the target. To address these issues, some studies have introduced affinity learning-based methods [21], [22], [60], which directly aggregate the similarity between support–query pixel pairs into the query features. For example, Zhang et al. [21] proposed a cyclic-consistent transformer for the FSS task, enhancing query features through self-attention and cross-attention mechanisms. Xu et al. [22] introduced a framework called SCCAN, which effectively integrates support samples and query features via a self-calibrated cross-attention (SCCA) module, addressing background mismatch and foreground–background entanglement issues.

Recently, some work has extended the FSS task to remote sensing scenarios [25], [61], [62], [63], [64], [65], aiming to address the labeling challenges in the remote sensing domain. For example, Lang et al. [25] designed a framework called $R^2$Net, specifically for remote sensing tasks. This framework employs global correction and decoupled registration to dynamically fuse global and local prototypes, significantly improving the performance of remote sensing FSS. Bi et al. proposed DMNet [62], which performs dual mining of shared semantics between support and query images as well as query-specific semantics, providing new insights for future work. Different from these methods, which rely on limited support samples to extract semantic information for guiding query segmentation, our MVLPNet introduces textual information to establish more robust segmentation guidance.

### D. Text Prompts in FSS Task

With the growing interest in integrating language priors into vision tasks and the success of vision-language models, such as CLIP, some works [31], [32], [66], [67], have focused on exploring the role of text prompts in FSS, achieving notable performance improvements. For example, PGMA-Net [66] unifies textual and visual cues within a class-agnostic framework, enabling the activation of generalizable mask prototypes and enhancing the generalization ability of the FSS model. PAT [67] innovatively proposes a dynamic class-aware prompting paradigm, which jointly optimizes the encoder through cross-modal semantic guidance from textual prompts and local mask generation, enabling it to focus more precisely on target objects and achieving excellent performance. In addition, Wang et al. [31] proposed a novel method, HSE, which effectively addresses significant intraclass variations in remote sensing images by embedding semantic knowledge from class descriptions into visual features. However, these methods simply fuse textual and visual features via concatenation or cosine similarity, overlooking the inherent limitations between the two modalities and failing to fully exploit the advantages of textual cues. In contrast, our TVCE module leverages the OT plan to achieve tight alignment between textual and visual

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features, providing more robust guidance for the segmentation of target regions.

## III. Methodology

### A. Task Description

Unlike traditional semantic segmentation, which segments only objects from known classes, FSS aims to segment objects from unseen classes using only a few support samples, without requiring additional training. Specifically, we construct a training dataset $D_{\text{train}}$ and a testing dataset $D_{\text{test}}$ based on two disjoint class sets, $C_{\text{base}}$ and $C_{\text{novel}}$, to align with the inference setup, thus improving the generalizability of the model. We follow the mainstream approach by employing a metalearning strategy to train the model, sampling a series of episodes from $D_{\text{train}}$ to simulate the few-shot setting during the metatesting phase. Each episode contains a support set $S = \{I_j^S, M_j^S\}_{j=1}^K$ and a query set $Q = \{I^Q, M^Q\}$, both belonging to the same semantic category $c$, where $I_j^*$ and $M_j^*$ represent the original image and its corresponding binary mask, respectively. During the metatraining phase, both the support and query sets are input into the FSS model to generate the predicted results $\tilde{M}_Q$. The query mask $M_Q$ is used to compute the loss during the metatraining phase, but it is unavailable during the metatesting phase. Importantly, $K$ denotes the number of support images, and based on previous work, we focus on evaluating FSS performance for $K = 1$ and $K = 5$.

### B. Method Overview

The proposed MVLPNet focuses on alleviating intraclass variations and suppressing interference from complex background regions in remote sensing images, and consists of the TVCE and PGSA modules. The overall architecture is shown in Fig. 2. Specifically, given the support–query image pair $I^S$ and $I^Q$, we follow the previous methods [59] and extract features from a weight-shared backbone network (VGG16 [68] or ResNet [69]) to obtain their mid-level features $\{F_Q, F_S\}$ (from blocks 2 and 3), as well as high-level features $\{F_Q^h, F_S^h\}$ (from block 4). In the initial feature extraction phase, we obtain the support foreground features $v_s$ and the query prior mask $M_{\text{prior}}$. Meanwhile, TVCE uses the CLIP model to extract visual features and category-specific text embeddings from the query image. Then, an OT plan is established to tightly align these embeddings with the visual features of the query image, thereby extracting semantic information from the query image to guide its segmentation. Next, the support mid-level features $F_S$ with the support mask and the support foreground features $\hat{v}_s$ are fed into the SIA to extract more comprehensive support information for precise matching. Then, PC is applied to capture the consistent foreground semantics between the support–query image pair ($F'_Q$ and $F_{\text{main}}$). Finally, the refined query features are input into the decoder to generate the final prediction.

### C. Initial Feature Extraction Stage

The core challenge of FSS lies in transforming the support image and its mask into class-specific guidance for segmenting the foreground in the query image. Inspired by Tian et al. [59], we use a prior mask to approximately localize the foreground region in the query image. Formally, given $F_Q^h \in \mathbb{R}^{C\times H\times W}$ and $F_S^h \in \mathbb{R}^{C\times H\times W}$, the prior mask of the query image can be defined as

$$M_q = B\left(\mathcal{G}\left(M_S, F_Q^h, F_S^h\right)\right) \tag{1}$$

where $M_q \in \mathbb{R}^{H\times W}$, $\mathcal{G}(.,.,.,)$ is the prior generator, $M_S$ denotes the support mask, and $B$ refers to the operation where $M_q$ is broadcasted, changing its shape from $H \times W$ to $1 \times H \times W$.

Next, to suppress irrelevant background information and extract class-specific foreground cues from the support features, we apply a MAP operation to compute the support prototype, which serves as guidance for segmenting the query image. Formally, given the mid-level features $F_S$ of the support image and the support mask $M_S$, the prototype is calculated as

$$v_s = B(\text{MAP}(F_S, M_S)) \tag{2}$$

where $v_s$ compresses all the support foreground pixel information, and $B$ represents that we use broadcast $v_s$ so that its shape changes from $C \times 1$ to $C \times H \times W$.

### D. Textual–Visual Consistency Enhancement

Most existing FSS methods [25], [30], [61], [62] rely primarily on support images to generate prototypes that guide segmentation. However, the limited support samples are often insufficient to capture the extreme intraclass variations in remote sensing images, leading to performance degradation. Inspired by PI-CLIP [32], we propose leveraging the powerful visual and textual understanding capabilities of CLIP to provide more robust guidance. However, directly aligning the features of the two modalities through cosine similarity introduces new challenges: CLIP focuses primarily on high-level associations between image and text features rather than fine-grained visual features [70], [71], [72]. Therefore, forced alignment between image and text features can result in incorrect activations. We further argue that this issue may be exacerbated by the inherent challenges of high interclass similarity in remote sensing images [73], [74].

To this end, we model the alignment of the two modalities as an OT problem. By searching for the transport plan that minimizes the transport cost, we optimize the mapping between the fine-grained image features and text features, thereby effectively extracting semantic information from the query image itself to guide segmentation. This approach can be efficiently solved via the Sinkhorn algorithm [75]. As shown in Fig. 2, the TVCE module first takes the query image with the class name as input: the query image is fed into the CLIP image encoder to generate the corresponding query features $F_i \in \mathbb{R}^{M\times D}$, where $M = H \times W$ corresponds to the product of the height $H$ and width $W$, and $D$ denotes the vector dimension. At the same time, we innovatively define a set of target class text prompts as guidance for the model. The target class text prompts are defined as "a photograph of target class," and the nontarget text prompts are defined as "a photograph

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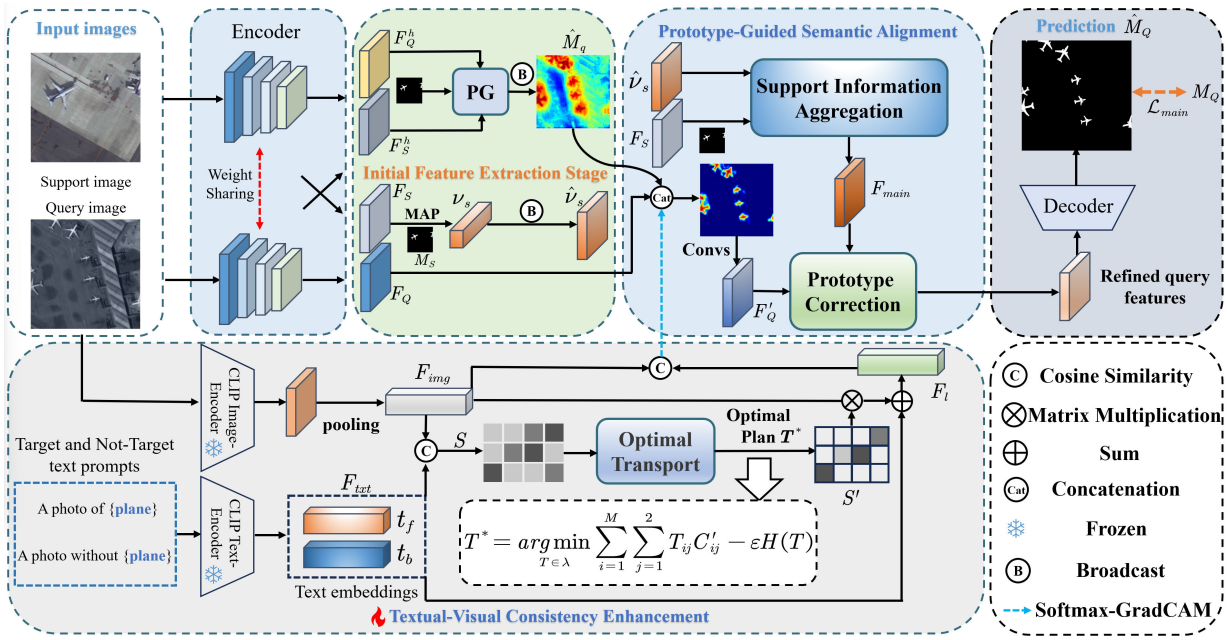



Fig. 2. Overall framework of the proposed MVLPNet method in the one-shot setting: it first extracts mid- and high-level query and support features, query prior masks $\hat{M}_q$, and support foreground features $\hat{v}_s$. Meanwhile, we input the query image with the target and nontarget text prompts into the TVCE module to extract category information from the query image itself, thus surpassing the limited support set to better guide the query segmentation. Next, we input the enhanced query features and support features into the PGSA module to explicitly guide the model to focus more on the consistent foreground regions between the query and support features rather than the background ones.

without target class." These are then input into the CLIP text encoder to generate two category-specific text embeddings $t_f \in \mathbb{R}^{1\times D}$ *and* $t_b \in \mathbb{R}^{1\times D}$, which are concatenated to form the final text embeddings $F_{\text{txt}} \in \mathbb{R}^{2\times D}$.

Next, the text–visual alignment score map can be represented as

$$S = \text{Cosine}\left(F_{\text{txt}}, F_{\text{img}}\right) \tag{3}$$

where Cosine$(\cdot)$ denotes the cosine similarity.

To ensure that the text embeddings align with the most relevant pixel distributions, we first compute the text–visual alignment score map $S \in \mathbb{R}^{2\times M}$ from 3, which represents the similarity matrix between the text embeddings and the query image features. Then, we define the cost matrix $C' = 1 - S$. Notably, the larger $C'(i, j)$ is, the higher the similarity between the $i$th query pixel and the $j$th text embedding, resulting in a lower transport cost. At this point, the transport plan is denoted as $T \in \mathbb{R}^{2\times M}$, where the plan aims to assign each of the $M$ image pixels to the two text embeddings. Therefore, the transport plan $T$ serves as the mapping matrix that maximizes the cosine similarity between embeddings from different modalities. The optimization function is as follows:

$$T^* = \underset{T\in\lambda}{\text{argmin}} \sum_{i=1}^{M}\sum_{j=1}^{2} T_{ij}C'_{ij} - \varepsilon H(T) \tag{4}$$

where $H(T) = \sum_{ij} T_{ij} \log T_{ij}$ and $\varepsilon > 0$ is the regularization parameter, and $T^*$ denotes the OT plan. Besides, the transport plan $T$ needs to obey the distribution of $\lambda$

$$\lambda = \left[T \in \mathbb{R}^{2\times M}, T1^M = \mu, (T)^{-1}1^2 = \nu\right] \tag{5}$$

where $1^2$ and $1^M$ denote vectors with values of 1 in all dimensions. For 4, we have an optimization solution when $t \le t_{\max}$ as follows:

$$T^* = \text{diag}\left(a^t\right)\exp\left(\frac{-C'}{\varepsilon}\right)\text{diag}\left(\exp\left(b^t\right)\right)$$
$$\begin{cases} a^t = \log\mu - \log\left[\sum \exp\left[-\frac{1}{\varepsilon}C' + b^{t-1}\right]\right] \\ b^t = \log\nu - \log\left[\sum \exp\left[-\frac{1}{\varepsilon}\left(C'\right)^T + a^{t-1}\right]\right] \end{cases} \tag{6}$$

where $t$ is the iteration, with $b^0$ initialized to 1.

After deriving the OT plan $T^*$, we use it to refine the score map $S$, which is used to determine which fine-grained features contribute most to the update of the text features. The process is as follows:

$$S' = S \odot T^* \tag{7}$$

where $\odot$ denotes the dot product operation.

Then, the weighted visual information is calculated using $S' \in \mathbb{R}^{2\times M}$ and visual features $F_{\text{img}}$ to update the text features $F_{\text{txt}}$ obtained from the CLIP text encoder. The process is as follows:

$$F_l = F_{\text{txt}} + \eta \times \left(S' \otimes F_{\text{img}}\right) \tag{8}$$

where $\otimes$ denotes the matrix multiplication and $\eta$ is the learnable parameter.

Next, we realign the updated text features $F_l$ and visual features $F_{\text{img}}$ and use Softmax operation to obtain classification scores

$$\omega_l = \text{Softmax}\left(\text{Cosine}\left(F_l, F_{\text{img}}\right)/\tau\right) \quad l \in \{f, b\} \tag{9}$$

where $\tau$ denotes the temperature parameter.

Finally, based on the resulting foreground classification scores $\omega_f$, Grad-CAM [33] is applied to generate the feature

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maps $F_{\text{prior}} \in \mathbb{R}^{1\times H\times W}$, thereby enhancing the features of the target regions

$$\mu_m = \frac{1}{HW}\sum_i\sum_j \frac{\partial \omega_f}{\partial F^m_{\text{img}}(i,j)}$$
$$F_{\text{prior}} = \text{ReLU}\left(\sum_m \mu_m F^m_{\text{img}}\right) \tag{10}$$

where $\mu_m$ denotes the weight for $m$th feature map of the foreground regions, $F^m_{\text{img}}$ represents the activation value of $m$th feature map, and $(i, j)$ means the pixel position.

### E. Prototype-Guided Semantic Alignment

Existing methods [24], [25], [29], [30] have focused on ambiguously matching query features using one or multiple prototypes for segmentation, overlooking interference from complex background regions in remote sensing images, which leads to incorrect activations. To address this issue, we propose a PGSA module, which consists of an SIA module and a PC module. PGSA aims to precisely match support and query features, enabling consistent activation of foreground regions while suppressing irrelevant background regions.

*1) Support Information Aggregation:* Specifically, we first concatenate the mid-level features $F_S$, support foreground features $\hat{v}_s$, and support mask $M_S$ as an input $F'_S \in \mathbb{R}^{C\times H\times W}$, which is used for segmenting the target regions of the query image. Meanwhile, we concatenate $\hat{M}_q$, $F_{\text{prior}}$ and $F_Q$ to construct another input $F'_Q \in \mathbb{R}^{C\times H\times W}$. Since $\hat{M}_q$, and $F_{\text{prior}}$ activate only the regions corresponding to the target, they effectively enhance the semantic clarity between the foreground and background in the query image, thus providing more reliable guidance for pixelwise matching between query and support features. The detailed process is as follows:

$$F'_S = \text{Convs}\big(\text{Cat}\big(F_S, \hat{v}_s, M_S\big)\big)$$
$$F'_Q = \text{Convs}\big(\text{Cat}\big(F_{\text{prior}}, \hat{M}_q, F_Q\big)\big) \tag{11}$$

where Convs denotes that the use of a $1 \times 1$ convolution to compress the channel dimension to $C$.

Since support foreground features are inevitably lost during feature extraction, directly utilizing support features for target region segmentation leads to incomplete activations. To address this issue, we propose the SIA module to aggregate the foreground regions lost in the support features, providing more accurate guidance. As shown in Fig. 3(a), an initial prediction is performed using the support features $F'_S$ and the support mask $M_S$ to obtain the predicted mask $M_{\text{aux}}$

$$F' = \text{Convs}\big(\text{Cat}\big(B\big(v'\big), F'_S\big)\big)$$
$$M_{\text{aux}} = \text{Argmax}\big(F'\big) \tag{12}$$

where $M_{\text{aux}}$ is the binary mask, with 0 indicating the background and 1 representing the class $c$.

Finally, using the predicted mask $M_{\text{aux}}$ and the ground-truth mask $M_S$, we can obtain the primary support vector $V_{\text{pri}}$ and the auxiliary support vector $V_{\text{aux}}$. These vectors are then aggregated to obtain $F_{\text{main}} \in \mathbb{R}^{C\times H\times W}$, thereby mining information

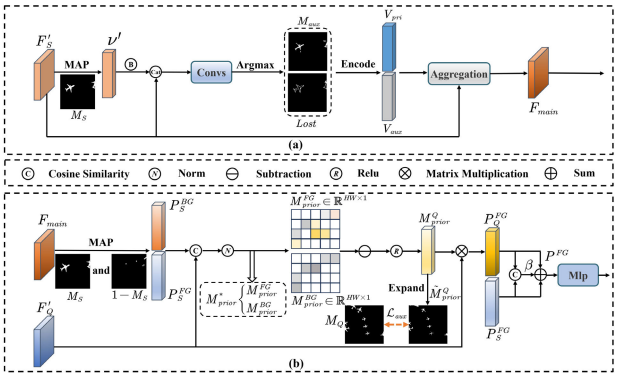



Fig. 3. Specific structure of (a) SIA and (b) PC modules.

that is prone to being lost and enabling more precise matching

$$\begin{cases} V_{\text{pri}} = \dfrac{\sum_{i=1}^{hw} F'(i)\,[M_S(i)=1]\,[M_{\text{aux}}(i)=1]}{\sum_{i=1}^{\text{hw}}[M_S(i)=1][M_{\text{aux}}(i)=1]} \\ V_{\text{aux}} = \dfrac{\sum_{i=1}^{\text{hw}} F'(i)\,[M_S(i)=1]\,\big[M_{\text{aux}}(i)\neq 1\big]}{\sum_{i=1}^{\text{hw}}[M_S(i)=1]\big[M_{\text{aux}}(i)\neq 1\big]} \end{cases} \tag{13}$$
$$F_{\text{main}} = \text{Convs}\big(\text{Cat}\big(F'_S, B\big(V_{\text{pri}}, V_{\text{aux}}\big)\big)\big) \tag{14}$$

where $[M_S(i) = 1] \cdot [M_{\text{aux}}(i) = 1]$ indicates the correctly predicted foreground mask, and $[M_S(i) = 1] \cdot [M_{\text{aux}}(i) \neq 1]$ denotes the missing foreground mask. $V_{\text{pri}}$ retains the primary support information and $V_{\text{aux}}$ focuses on collecting the missing key information.

*2) Prototype Correction:* Fig. 3(b) illustrates the flowchart of PC. Given the support feature $F_{\text{main}}$ and support mask $M_S$, we first apply MAP to extract the support foreground and background feature prototypes

$$P^{\text{FG}}_S = \text{MAP}(F_{\text{main}}, M_S)$$
$$P^{\text{BG}}_S = \text{MAP}(F_{\text{main}}, 1 - M_S). \tag{15}$$

Then, the cosine similarity between the query features and both the support foreground and background prototypes is computed to obtain the query prior masks ($M^{\text{FG}}_{\text{prior}}$ and $M^{\text{BG}}_{\text{prior}}$). Subsequently, these prior masks are normalized to obtain $M^*_{\text{prior}}$. This procedure can be formulated as follows:

$$M^*_{\text{prior}} = \text{Norm}\big(\text{Cosine}\big(F'_Q, P^*_S\big)\big) \tag{16}$$

where $* \in \{FG, BG\}$, Norm$(\cdot)$ refers to the min–max scaler that normalizes the values into the range [0,1].

To remove ambiguous query regions that exhibit high similarity to both support foreground and background features, we perform a clipped subtraction to rectify and obtain the query foreground pseudo-mask $M^Q_{\text{prior}}$

$$M^Q_{\text{prior}} = \text{Relu}\left(M^{\text{FG}}_{\text{prior}} - M^{\text{BG}}_{\text{prior}}\right) \tag{17}$$

where Relu$(\cdot)$ is the operator that sets negative values to zero. These negative values correspond to query regions that are similar to the support background features.

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Next, a matrix multiplication is performed to aggregate the query foreground pseudo-mask $M^{Q}_{\text{prior}}$ into the most discriminative query foreground prototype $P^{\text{FG}}_{Q}$

$$P^{\text{FG}}_{Q} = \text{Softmax}\left(M^{Q}_{\text{prior}}\right) \otimes F'_{Q}. \tag{18}$$

After that, the cosine similarity $\beta$ between $P^{\text{FG}}_{S}$ and $P^{\text{FG}}_{Q}$ is calculated. $\beta$ is utilized to weight and fuse the prototypes based on their importance and relevance to obtain $P^{\text{FG}}$. The process is as follows:

$$\begin{aligned} \beta &= \text{Cosine}\left(P^{\text{FG}}_{S}, P^{\text{FG}}_{Q}\right) \\ P^{\text{FG}} &= \beta \cdot P^{\text{FG}}_{S} + (1-\beta) \cdot P^{\text{FG}}_{Q} \end{aligned} \tag{19}$$

where $P^{\text{FG}}$ contains both the support foreground prototype $P^{\text{FG}}_{S}$ and the most discriminative query foreground prototype $P^{\text{FG}}_{Q}$. Finally, $P^{\text{FG}}$ is processed by a multilayer perceptron (MLP) for feature refinement. Guided by the most discriminative query foreground prototype, the PC module can adaptively aggregate regions where the support and query features are consistent, achieving more accurate segmentation.

### F. Training Losses

To better optimize the model parameters during the training phase, we use cross entropy (CE) as the primary loss function to evaluate the difference between the predicted mask $\tilde{M}_{Q}$ of the query image and the ground truth mask $M_{Q}$

$$\mathcal{L}_{m} = \text{CE}\left(M_{Q}, \tilde{M}_{Q}\right). \tag{20}$$

In addition, we also use binary cross entropy (BCE) to further regularize the quality of the generated query foreground pseudo-mask $P^{Q}_{\text{prior}}$, achieving balanced optimization

$$\begin{aligned} \tilde{M}^{Q}_{\text{prior}} &= \text{Expand}\left(M^{Q}_{\text{prior}}\right) \\ \mathcal{L}_{\text{aux}} &= \text{BCE}\left(M_{Q}, \tilde{M}^{Q}_{\text{prior}}\right). \end{aligned} \tag{21}$$

Finally, the model's parameters are updated by simultaneously optimizing all the previously mentioned losses

$$\mathcal{L} = \mathcal{L}_{m} + \alpha \mathcal{L}_{\text{aux}} \tag{22}$$

where $\alpha$ is the hyperparameter for balancing corresponding losses.

## IV. Experiments

In this section, we experimentally validate the effectiveness of the proposed method. First, we introduce the datasets, evaluation metrics, and implementation details used in the experiments. Next, we compare the performance of our model with that of state-of-the-art methods. We then conduct ablation studies to assess the contribution of each proposed module. Finally, we perform visualization experiments to qualitatively analyze the importance of each module.

### A. Experimental Setup

*1) Datasets:* To ensure a fairer comparison, we follow previous work [25] and use the iSAID-5i dataset [76], which contains 655 451 object instances across 15 geospatial categories from 2806 high-resolution images. Specifically, in the few-shot setting, these geospatial categories are evenly divided into three folds: $5^{i}$: $i \in \{0, 1, 2\}$. The model is trained on two folds (ten object categories) and tested on the remaining fold (five object categories). For each fold, we randomly select 1000 support–query image pairs for performance evaluation during the testing phase.

The LoveDA dataset [77] contains 5987 remote sensing images and 166 768 object instances, with each image having a resolution of 1024 × 1024. It includes seven categories: background, buildings, roads, and others. On this basis, Bi et al. [62] created the LoveDA-2i benchmark. Following this work, we exclude the background category and use only the remaining six categories. Similar to the iSAID-5i dataset, we divide the six categories of the LoveDA dataset into three parts.

*2) Evaluation Metrics:* In this article, we use the mean intersection over union (mIoU) and foreground–background intersection over union (FB-IoU) as our evaluation metrics. The mIoU computes the average IoU score across all foreground classes in a fold, providing a comprehensive measure of segmentation accuracy. The formula is given by: $\text{mIoU} = (1/n)\sum_{c=1}^{N_c} \text{IoU}_{c}$, where $n$ denotes the number of new classes, and $\text{IoU}_{c}$ represents the IoU score for class $c$. The latter is the result obtained by averaging the foreground IoU and background IoU, the calculation is expressed as: $\text{FB-IoU} = (\text{IoU}_{F} + \text{IoU}_{B})/2$, where $\text{IoU}_{F}$ and $\text{IoU}_{B}$ indicate the foreground and background IoU scores, respectively.

### B. Implementation Details

The model is implemented within the PyTorch framework. Following previous work [25], we train a segmentation network on the base dataset via a standard learning paradigm, which provides a more suitable backbone for FSS in remote sensing images. The learning rate is set to $1.25e^{-3}$, the batch size is 16, and the model is trained for 50 epochs. The training data in this phase aligns with the metatraining data used in the few-shot model. To ensure a fair comparison, we use this backbone network to replicate all the comparison methods. During the metatraining phase for FSS, the parameters of the backbone network are frozen. For the iSAID-5i dataset, the input image sizes are 512 × 512 and 256 × 256, the batch size is 16, and the learning rate is 0.005. For LoveDA-2i, the input image resolution is 473 × 473, the batch size is 16, and the learning rate is 0.0015. Data augmentation techniques, such as scaling, rotation, and Gaussian blur, are employed to increase data diversity and improve the model's generalizability. The number of training epochs is set to 25 for both datasets. All experiments are conducted on a single NVIDIA V100 GPU.

The baseline of our model consists of PFENet [59] and ASPP [37], where PFENet provides prior information to roughly localize the foreground in the query image, while

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TABLE I

PRECISION RESULTS OF THE MODEL ON THE iSAID-5i (512 × 512) BENCHMARK UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ARE PRESENTED BELOW. THE EVALUATION METRICS INCLUDE mIoU FOR EACH FOLD, THE MEAN mIoU ACROSS ALL FOLDS (MEAN), AND FB-IoU. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

| Backbone | Method | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) |
| VGG-16 | PANet(ICCV2019) [58] | 26.86 | 14.56 | 20.69 | 20.70 | 52.37 | 30.89 | 16.63 | 24.05 | 23.86 | 53.49 |
| | CANet(CVPR2019) [78] | 13.91 | 12.94 | 13.67 | 13.51 | 51.83 | 17.32 | 15.07 | 18.23 | 16.87 | 52.25 |
| | PFENet(TPAMI2020) [59] | 28.52 | 17.05 | 18.94 | 21.50 | 54.72 | 37.59 | 23.22 | 30.45 | 30.42 | 57.23 |
| | BAM(CVPR2022) [79] | 33.93 | 16.88 | 21.47 | 24.09 | 57.62 | 38.46 | 22.76 | 28.81 | 30.01 | 61.14 |
| | $R^2$Net(TGRS2023) [25] | 35.27 | 19.93 | 24.63 | 26.61 | 60.37 | **42.06** | 23.52 | 30.06 | 31.88 | 62.67 |
| | DMNet(TGRS2023) [62] | **38.42** | 25.36 | 34.53 | 32.77 | 52.36 | 41.32 | 27.41 | 37.18 | 35.30 | 53.24 |
| | $MS^2A^2$Net(TGRS2023) [30] | 32.46 | 23.15 | 30.76 | 28.79 | 59.17 | 38.56 | 25.23 | 33.57 | 32.45 | 61.41 |
| | PI-CLIP(CVPR2024) [32] | 34.27 | 22.56 | 32.16 | 29.66 | 53.39 | 37.24 | 26.03 | 35.26 | 32.84 | 55.74 |
| | MGANet(TGRS2024) [61] | 37.21 | **27.45** | 35.24 | 33.30 | 52.71 | 40.47 | 28.54 | 38.47 | 35.83 | 54.26 |
| | **Ours** | 38.10 | 26.72 | **36.42** | **33.75** | **63.89** | 39.57 | **29.74** | **40.89** | **36.73** | **66.20** |
| ResNet-50 | PANet(ICCV2019) [58] | 27.56 | 17.23 | 24.60 | 23.13 | 56.56 | 36.54 | 16.05 | 26.22 | 26.27 | 57.37 |
| | CANet(CVPR2019) [78] | 25.51 | 13.50 | 24.45 | 21.15 | 56.64 | 29.32 | 21.85 | 26.91 | 26.03 | 59.46 |
| | PFENet(TPAMI2020) [59] | 35.84 | 23.35 | 27.20 | 28.80 | 60.09 | 42.42 | 25.34 | 33.00 | 33.59 | 63.25 |
| | BAM(CVPR2022) [79] | 39.43 | 21.69 | 28.64 | 29.92 | 62.04 | 43.29 | 27.92 | 38.62 | 36.61 | 65.00 |
| | $R^2$Net(TGRS2023) [25] | 41.22 | 21.64 | 35.28 | 32.71 | 63.86 | **46.45** | 25.80 | 39.84 | 37.36 | 66.18 |
| | DMNet(TGRS2023) [62] | 40.72 | 26.47 | 40.37 | 35.85 | 53.14 | 41.83 | 28.34 | 42.56 | 37.58 | 55.72 |
| | $MS^2A^2$Net(TGRS2023) [30] | 34.96 | 25.30 | 30.71 | 30.32 | 59.81 | 42.30 | 27.10 | 36.31 | 35.24 | 62.01 |
| | PI-CLIP(CVPR2024) [32] | 35.54 | 26.13 | 37.63 | 33.10 | 56.14 | 41.36 | 27.35 | 39.64 | 36.12 | 58.32 |
| | MGANet(TGRS2024) [61] | 39.90 | **30.72** | **44.18** | **38.27** | 54.32 | 42.35 | **31.58** | 46.37 | 40.10 | 56.19 |
| | **Ours** | **41.91** | 28.27 | 41.03 | 37.07 | **65.23** | 44.68 | 30.70 | **49.01** | **41.46** | **68.73** |

ASPP enhances adaptability to the multiscale characteristics of remote sensing scenes.

### C. Comparison With State of the Arts

In this section, we compare our method with existing FSS approaches on two widely used remote sensing FSS datasets, including both methods designed for natural scenes and those specifically tailored for remote sensing.

*1) iSAID-5i (512 × 512):* The performance of our method on the iSAID-5i dataset under one-shot and five-shot settings is shown in Table I. As indicated in Table I, our method achieves state-of-the-art results compared with all other methods. Specifically, in the one-shot setting, the mean mIoU with VGG-16 improves by 0.45% compared with MGANet [61], while with ResNet-50 as the backbone, the mIoU is 1.20% lower than that of MGANet [61]. However, our method effectively distinguishes foreground and background features by extracting the most discriminative query foreground prototypes and achieves 63.89% and 66.20% in FB-IoU for the one-shot and five-shot settings, respectively, significantly outperforming the other methods. Notably, methods that perform well in natural scenes, such as PI-CLIP [32] and PFENet [59], fail to achieve similar results in remote sensing scenes. This is due to the extreme intraclass variability and complex background interference present in remote sensing images, which are less common in natural images. Our method effectively addresses these challenges. In the five-shot setting, our method continues to outperform the previous SOTA method MGANet [61]. Specifically, the mean mIoU increases by 0.90% with VGG-16 and 1.36% with ResNet-50. Meanwhile, our method maintains superior performance in FB-mIoU. This fully demonstrates that, compared with other methods, the integration of text prompts is beneficial for establishing robust class-specific segmentation guidance.

In addition, we present the classwise performance of all methods under the one-shot setting, as shown in Table II. From left to right, the 15 geospatial classes (C1–C15) represent the segmentation accuracy of each novel class across the folds. Our approach with the ResNet-50 backbone achieves the top-1 results in 6 out of 15 cases. Specifically, our method outperforms in categories such as ship (C1), storage tank (C2), tennis court (C4), roundabout (C12), and plane (C14). However, it faces challenges in segmenting small vehicle (C9) and harbor (C15). This is primarily due to interference from objects with similar appearances across classes, such as the similarity between large vehicle (C8) and small vehicle (C9), which makes it difficult to distinguish their corresponding class prototypes based solely on scale differences. For the harbor (C15) class, the high density of close ships and dock facilities increases the segmentation difficulty. In contrast, categories such as plane (C14) and ship (C1) typically exhibit more distinctive features and are less influenced by irrelevant interference, resulting in higher segmentation accuracy.

*2) iSAID-5i (256 × 256):* To further demonstrate the effectiveness of our method, we follow the input image size (256 × 256) setting in [67] and conduct more detailed qualitative and quantitative comparisons with recent text-prompt-based approaches [32], [66], [67] on the iSAID-5i dataset. Table III presents the one-shot and five-shot results. Specifically, our method achieves 51.80% mean mIoU and 67.34% FB-IoU under the one-shot setting with ResNet-50 as the backbone, outperforming the best transformer-based competitor [67] by 1.29% and 2.87%, respectively. Notably, while PGMA-Net [66] performs well in natural image scenarios, it fails to achieve satisfactory results in remote sensing applications. This is because CLIP focuses primarily on high-level associations between image and text features. When cosine similarity is directly used to align these features, the lack of fine-grained semantic alignment limits the model's ability to cope with the extreme intraclass variations in remote sensing images. In addition, in the five-shot setting, our method achieves the mIoU of 56.09%, which is 3.32% lower than that of the best-performing method, PAT [67]. This is mainly because, under the one-shot setting with only a single support sample,

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TABLE II

CLASSWISE PERFORMANCE COMPARISON IN MIOU UNDER ONE-SHOT SETTING USING RESNET-50 BACKBONE ON THE ISAID-5I DATASET

| Method | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12 | C13 | C14 | C15 | Mean(%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | ResNet-50 | | | | | | | | | | | | | | | |
| PANet [58] | 21.81 | 36.31 | 23.01 | 42.06 | 14.59 | 12.11 | 17.44 | 22.70 | 12.27 | 21.60 | 30.29 | 24.62 | 26.79 | 25.54 | 15.79 | 23.13 |
| CANet [78] | 39.57 | 18.54 | 18.46 | 33.63 | 17.34 | 9.78 | 5.49 | 22.15 | 5.17 | 24.89 | 9.96 | 36.50 | 19.12 | 38.82 | 17.85 | 21.15 |
| PFENet [59] | 39.02 | 45.63 | 20.86 | 49.96 | 23.72 | 21.00 | 24.76 | 31.59 | 6.98 | 32.42 | 13.34 | 47.64 | 30.65 | 32.82 | 11.54 | 28.80 |
| BAM [79] | 36.34 | 39.76 | **38.23** | 58.13 | 24.71 | 18.25 | 12.68 | 35.91 | 11.42 | 30.21 | 28.98 | 40.74 | 29.43 | 33.25 | 10.79 | 29.92 |
| $R^2$Net [25] | **46.87** | 49.06 | 30.70 | 52.86 | 26.62 | 24.31 | 17.25 | 31.25 | 13.67 | 21.73 | 24.88 | 46.07 | 42.29 | 42.07 | 21.08 | 32.71 |
| DMNet [62] | 41.57 | 47.28 | 36.68 | 52.47 | 25.60 | 26.91 | 30.44 | 30.35 | 16.53 | 28.12 | **40.73** | 58.69 | 43.76 | 43.51 | 15.16 | 35.85 |
| $MS^2A^2$Net [30] | 38.27 | 43.44 | 25.40 | 40.13 | **27.56** | 23.18 | 18.92 | 33.68 | 15.47 | 35.25 | 26.79 | 38.36 | 33.73 | 25.62 | **29.05** | 30.32 |
| PI-CLIP [32] | 37.25 | 44.50 | 26.94 | 46.18 | 22.83 | 26.05 | 25.74 | 35.48 | 17.23 | 26.15 | 25.24 | 52.73 | **47.36** | 42.18 | 20.64 | 33.10 |
| MGANet [61] | 42.35 | 48.15 | 29.20 | 54.37 | 25.43 | **32.75** | 29.11 | **36.08** | 17.69 | **37.97** | 36.98 | 62.17 | 45.82 | 47.36 | 28.57 | **38.27** |
| **Ours** | 40.98 | **50.40** | 34.73 | **61.64** | 21.79 | 29.23 | **32.85** | 34.83 | **18.23** | 26.23 | 37.97 | **63.95** | 36.31 | **49.67** | 17.25 | 37.07 |

TABLE III

PRECISION RESULTS OF THE MODEL ON THE ISAID-5I (256 × 256) BENCHMARK UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ARE PRESENTED BELOW. THE EVALUATION METRICS INCLUDE MIOU FOR EACH FOLD, THE MEAN MIOU ACROSS ALL FOLDS (MEAN), AND FB-IOU. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

| Backbone | Method | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) |
| CLIP-RN50 | PGMA-Net(TMM2024) [66] | 50.34 | 39.27 | 51.52 | 47.04 | 59.69 | 54.23 | 42.57 | 55.41 | 50.74 | 63.17 |
| CLIP-RN101 | PGMA-Net(TMM2024) [66] | 53.19 | 42.94 | 53.01 | 49.71 | 61.85 | 55.10 | 47.68 | 58.26 | 53.68 | 67.06 |
| ViT-B/16 | AgMTR(IJCV2024) [65] | 55.41 | 40.67 | 53.81 | 49.98 | 63.20 | 59.05 | 47.86 | 64.08 | 57.00 | 67.60 |
| DeiT-B/16 | AgMTR(IJCV2024) [65] | **57.51** | 43.54 | 53.69 | 51.58 | 64.63 | 61.51 | 49.23 | 64.05 | 58.26 | 68.72 |
| DeiT-B/16 | PAT(TPAMI2024) [67] | 54.81 | 43.03 | 53.70 | 50.51 | 64.47 | **62.40** | **49.72** | **66.10** | **59.41** | 70.20 |
| VGG-16 | PI-CLIP(CVPR2024) [32] | 49.27 | 38.47 | 50.18 | 45.97 | 60.23 | 53.18 | 41.79 | 55.27 | 50.08 | 63.26 |
| | **Ours** | 52.02 | 42.63 | 51.97 | 48.87 | 64.69 | 56.94 | 45.34 | 61.28 | 54.52 | 66.75 |
| ResNet-50 | PI-CLIP(CVPR2024) [32] | 52.84 | 42.51 | 52.35 | 49.23 | 62.03 | 55.24 | 44.06 | 56.15 | 51.82 | 65.34 |
| | **Ours** | 56.44 | **44.69** | **54.28** | **51.80** | **67.34** | 60.31 | 47.25 | 60.72 | 56.09 | **70.45** |

our TVCE module is capable of integrating CLIP with the OT plan to mine additional visual cues directly from the query image to guide the segmentation, thereby achieving significant performance improvements. However, in the five-shot setting, the performance improvement is less pronounced, while PAT [67] shows a clear advantage by dynamically tuning the encoder with class-aware prompts. By iteratively transferring fine-grained semantics from both support and query images into the prompts, their class-awareness is progressively enhanced, effectively guiding the encoder to generate more discriminative features and achieve accurate segmentation. Nevertheless, our method continues to achieve the highest FB-IoU and offers new insights for leveraging multimodal information in remote sensing FSS.

*3) LoveDA-2i:* Table IV presents the one-shot and five-shot results on the LoveDA-2i dataset. Our method consistently achieves the highest mean mIoU under both the one-shot and five-shot settings across all backbone networks. Specifically, in the one-shot setting, compared with the SOTA method MGANet [61], our method with the VGG-16 backbone achieves a 1.13% increase in mean mIoU, while with ResNet-50, the mIoU improves by 0.37%. It can also be observed that in the five-shot setting, our method achieves an improvement in mIoU of 0.34% with VGG-16 and 1.12% with ResNet-50 compared with MGANet [61]. The underlying reason can be attributed to the more complex background regions in the LoveDA-2i dataset, where previous methods, such as MGANet [61] and DMNet [62], focus on ambiguously matching query features using one or multiple prototypes for segmentation, overlooking interference from background regions, which leads to performance degradation. In contrast, our method can extract more comprehensive support information and precisely match the query features with the support features, effectively suppressing irrelevant background regions in the query image. Furthermore, we present the classwise performance of all methods under the one-shot setting, as shown in Table V. From left to right, the six geospatial classes (C1–C6) represent the segmentation accuracy for each novel class across folds. It can be seen that our approach with the ResNet-50 backbone achieves the top-1 results in three out of six cases.

*4) Model Complexity:* Model complexity is also an important dimension for evaluating model performance. Therefore, we conduct a comparative experiment on model complexity. As shown in Table VI, we compare the FB-IoU, the number of learnable parameters, the total number of parameters, and the inference time under ResNet-50 in both the one-shot and five-shot settings on the iSAID-5i dataset. Typically, the number of learnable parameters (MB) is calculated by summing all learnable weights and biases, while the total number of parameters (MB) refers to the sum of all parameters in the model. In addition, inference speed (frames/s) is achieved by calculating the number of images segmented per second by the model in the one-shot setting. As shown in Table VI, since the pretraining-CLIP model is frozen, our model has only 8.9M learnable parameters, which is relatively low compared with the other methods. However, it has the highest total number of parameters (205.1M), primarily due to the integration of the TVCE module, which uses a pretraining-CLIP model to extract visual features and category-specific text embeddings from the query image. An OT plan is then established to tightly align these embeddings with the visual features of the query image, which further extracts semantic information from the

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TABLE IV

PRECISION RESULTS OF THE MODEL ON THE LOVEDA-2I BENCHMARK UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ARE PRESENTED BELOW. THE EVALUATION METRICS INCLUDE MIOU FOR EACH FOLD, THE MEAN MIOU ACROSS ALL FOLDS (MEAN), AND FB-IOU. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

| Backbone | Method | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) | Fold-0 | Fold-1 | Fold-2 | Mean(%) | FB-IoU(%) |
| VGG-16 | PANet(ICCV2019) [58] | 13.68 | 11.80 | 8.74 | 11.41 | 24.35 | 19.51 | 15.34 | 15.32 | 16.72 | 26.57 |
| | CANet(CVPR2019) [78] | 14.58 | 17.35 | 13.12 | 15.02 | 26.15 | 16.26 | 18.24 | 16.57 | 17.02 | 27.82 |
| | PFENet(TPAMI2020) [59] | 18.07 | 24.56 | 32.07 | 24.90 | 35.66 | 20.36 | 27.51 | 33.26 | 27.04 | 40.24 |
| | BAM(CVPR2022) [79] | 17.12 | 21.37 | 33.68 | 24.06 | 38.94 | 18.21 | 24.51 | 34.32 | 25.68 | 42.59 |
| | $R^2$Net(TGRS2023) [25] | 23.25 | 26.61 | 32.49 | 27.45 | 45.58 | 29.57 | 33.01 | 36.63 | 33.07 | 48.37 |
| | DMNet(TGRS2023) [62] | 29.64 | 32.36 | 35.42 | 32.47 | 52.97 | 32.34 | 39.48 | 38.92 | 36.91 | 55.33 |
| | $MS^2A^2$Net(TGRS2023) [30] | 26.18 | 28.45 | 31.22 | 28.62 | **57.72** | 29.15 | 34.52 | 34.03 | 32.57 | 58.63 |
| | PI-CLIP(CVPR2024) [32] | 24.61 | 30.52 | 34.53 | 29.89 | 51.39 | 28.62 | 37.65 | 36.17 | 34.15 | 53.42 |
| | MGANet(TGRS2024) [61] | 28.93 | **35.16** | 37.98 | 34.02 | 54.47 | 33.02 | **42.62** | 41.26 | 38.97 | 56.67 |
| | **Ours** | **32.26** | 33.45 | **39.75** | **35.15** | 55.84 | **34.05** | 38.97 | **44.92** | **39.31** | **59.56** |
| ResNet-50 | PANet(ICCV2019) [58] | 17.99 | 14.42 | 13.25 | 15.22 | 26.75 | 21.67 | 19.26 | 19.68 | 20.20 | 29.77 |
| | CANet(CVPR2019) [78] | 16.83 | 19.33 | 16.32 | 17.49 | 27.23 | 18.01 | 20.41 | 17.58 | 18.67 | 35.76 |
| | PFENet(TPAMI2020) [59] | 19.44 | 23.71 | 28.46 | 23.87 | 36.47 | 20.58 | 25.15 | 30.17 | 25.30 | 40.26 |
| | BAM(CVPR2022) [79] | 19.35 | 22.26 | 32.69 | 24.77 | 40.61 | 22.26 | 25.18 | 31.49 | 26.31 | 43.98 |
| | $R^2$Net(TGRS2023) [25] | 24.19 | 27.44 | 35.91 | 29.18 | 46.74 | 30.15 | 34.66 | 37.13 | 33.98 | 50.43 |
| | DMNet(TGRS2023) [62] | 31.89 | 34.45 | 36.37 | 34.24 | 54.67 | 34.11 | 42.54 | 41.03 | 39.23 | 56.33 |
| | $MS^2A^2$Net(TGRS2023) [30] | 27.55 | 30.25 | 33.26 | 30.35 | 56.83 | 29.66 | 35.14 | 36.23 | 33.68 | 59.26 |
| | PI-CLIP(CVPR2024) [32] | 26.38 | 32.67 | 35.77 | 31.61 | 52.08 | 29.13 | 38.22 | 38.29 | 35.21 | 54.17 |
| | MGANet(TGRS2024) [61] | **33.75** | 37.26 | 39.51 | 36.84 | 55.46 | **36.25** | 43.47 | 42.36 | 40.69 | 57.24 |
| | **Ours** | 33.05 | **38.13** | **40.44** | **37.21** | **57.85** | 35.26 | **44.54** | **45.64** | **41.81** | **61.71** |

TABLE V

CLASSWISE PERFORMANCE COMPARISON IN MIOU UNDER ONE-SHOT SETTING USING RESNET-50 BACKBONE ON THE LOVEDA-2I DATASET

| Method | C1 | C2 | C3 | C4 | C5 | C6 | Mean(%) |
|---|---|---|---|---|---|---|---|
| | ResNet-50 | | | | | | |
| PANet [58] | 14.59 | 21.39 | 17.00 | 11.84 | 11.37 | 15.13 | 15.22 |
| CANet [78] | 13.52 | 20.14 | 25.60 | 13.06 | 13.28 | 19.36 | 17.49 |
| PFENet [59] | 21.54 | 17.34 | 26.94 | 20.48 | 23.71 | 33.21 | 23.87 |
| BAM [79] | 20.57 | 18.13 | 21.27 | 23.25 | 25.74 | 39.64 | 24.77 |
| $R^2$Net [25] | 19.58 | 28.80 | 31.74 | 23.14 | 25.38 | 46.44 | 29.18 |
| DMNet [62] | 32.35 | **31.43** | 42.29 | 26.61 | 26.04 | 46.70 | 34.24 |
| $MS^2A^2$Net [30] | 24.49 | 30.61 | 35.72 | 24.78 | 25.34 | 41.18 | 30.35 |
| PI-CLIP [32] | 30.49 | 22.27 | 37.06 | 28.28 | 24.67 | 46.87 | 31.61 |
| MGANet [61] | **39.49** | 28.01 | 46.18 | **28.34** | 24.43 | 54.59 | 36.84 |
| **Ours** | 37.65 | 28.44 | **48.45** | 27.80 | **26.15** | **54.72** | **37.21** |

TABLE VI

COMPARATIVE STUDY OF MODEL COMPLEXITY. L-PARAMS AND PARAMS DENOTE THE NUMBER OF LEARNABLE PARAMETERS AND TOTAL PARAMETERS. INFERENCE SPEED DENOTES THE NUMBER OF IMAGES THE MODEL CAN INFER PER SECOND

| Method | Multimodal | 1-shot | 5-shot | L-Params | Params | Inference Speed |
|---|---|---|---|---|---|---|
| PANet [58] | N | 56.56 | 57.37 | 23.6M | 23.6M | 20.2 |
| CANet [78] | N | 56.64 | 59.46 | 22.3M | **22.9M** | **20.8** |
| PFENet [59] | N | 60.09 | 63.25 | 10.8M | 53.4M | 15.7 |
| BAM [79] | N | 62.04 | 65.00 | **4.9M** | 51.7M | 19.5 |
| $R^2$Net [25] | N | 63.86 | 66.18 | 5.0M | 51.8M | 18.1 |
| DMNet [62] | N | 53.14 | 55.72 | 7.4M | 69.8M | 11.9 |
| $MS^2A^2$Net [30] | N | 59.81 | 62.01 | 12.5M | 85.2M | 11.3 |
| MGANet [61] | N | 54.32 | 56.19 | 10.1M | 83.5M | 10.3 |
| PI-CLIP [32] | Y | 56.14 | 58.32 | 14.5M | 195.6M | 6.2 |
| **Ours** | Y | **65.23** | **68.73** | 8.9M | 205.1M | 7.4 |

query image itself. Although a certain number of additional parameters are introduced, our model achieves the best FB-IoU results, reaching 65.23% and 68.73% under the one-shot and five-shot settings, respectively.

Notably, compared with the PI-CLIP method [32], which also utilizes multimodal features, our model is specifically designed for remote sensing FSS and achieves the highest FB-IoU. Meanwhile, our method achieves an inference speed of 7.4 frames/s, 1.2 frames/s faster than the PI-CLIP method. These results demonstrate the effectiveness of our method in integrating multimodal information into remote sensing tasks and provide new insights for the future development of this field.

### D. Ablation Studies

We conduct ablation experiments on the iSAID-5i dataset using the ResNet-50 backbone to assess the impact of each module on segmentation performance. To qualitatively analyze the contribution of each module, we visualize the activation maps, further demonstrating the effectiveness of the proposed modules.

*1) Effects of Modules:* As shown in Table VII, we conduct ablation experiments on the TVCE and PGSA modules. The results indicate that both the TVCE and PGSA modules lead to improvements over the baseline. Specifically, with only the PGSA module, the mIoU improves by 2.01%, and it introduces minimal computational cost, achieving an inference speed of 16.7 frames/s. This validates the feasibility of utilizing the most discriminative query foreground prototype to activate the query foreground regions most similar to the support foreground. Meanwhile, the model with only the TVCE module achieves a 2.42% increase in the mIoU, which indicates that the powerful text and visual feature understanding ability of the CLIP model helps to extract semantic information from the query image itself to guide segmentation, thereby improving segmentation accuracy. However, the TVCE module introduces a significant computational cost, primarily due to the need to leverage the pretrained CLIP model to extract both visual and textual features from the query image, resulting in additional computational complexity. When both modules are integrated, the model's mIoU increases by 4.92% and the FB-IoU improves by 2.99%. This demonstrates that our modules synergistically enhance each other, unleashing their full potential.

*2) Effects of PGSA Module:* To validate the effectiveness of the PGSA module, we conduct ablation experiments on the PGSA modules, as shown in Table VII, the mIoU and

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TABLE VII

EFFECTS OF THE TVCE AND PGSA MODULES UNDER ONE-SHOT SETTING ON iSAID-5i. L-PARAMS AND PARAMS DENOTE THE NUMBER OF LEARNABLE PARAMETERS AND TOTAL PARAMETERS. INFERENCE SPEED DENOTES THE NUMBER OF IMAGES THE MODEL CAN INFER PER SECOND. THE GFLOPS INDICATE THE FLOATING POINT OPERATIONS OF THE MODEL. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREE FOLDS COMPARED WITH THE BASELINE

| TVCE | PGSA | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU | L-Params | Params | GFLOPs | Inference Speed | △ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 39.23 | 23.64 | 33.57 | 32.15 | 62.24 | **4.6M** | **51.7M** | **73.1** | **18.6** | - |
| | ✓ | 40.19 | 25.81 | 36.49 | 34.16 | 63.20 | 7.8M | 56.6M | 125.2 | 16.7 | +2.01 |
| ✓ | | 40.73 | 26.94 | 36.05 | 34.57 | 64.19 | 5.7M | 200.2M | 429.5 | 9.8 | +2.42 |
| ✓ | ✓ | **41.91** | **28.27** | **41.03** | **37.07** | **65.23** | 8.9M | 205.1M | 481.6 | 7.4 | +4.92 |

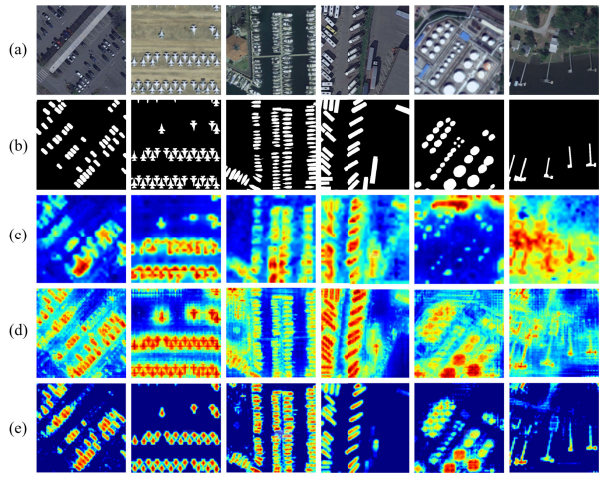



Fig. 4. Visualization results of different modules compared with the baseline. (a) Query image. (b) Query ground truth mask. (c) Activation map with the baseline. (d) Activation map result with the PGSA module. (e) Activation map result with the TVCE module.

FB-IoU are improved by 2.01% and 0.96%, respectively. We also present several visualization results in Fig. 4(d). Compared with baseline, incorporating this module results in clearer activation of target regions in the query image while suppressing complex background. For example, in the first column of the figure, the features of the "small vehicle" region are enhanced, and the boundaries are made clearer after incorporating the PGSA module, demonstrating its ability to activate areas most relevant to the foreground of the support image. In the second column, the baseline fails to fully activate the "plane" region and produces blurred boundaries. After incorporating the PGSA module, the target region becomes more prominent and is clearly separated from the surrounding background features.

In addition, we decompose the PGSA module into the SIA and PC modules for ablation experiments. As shown in Table VIII, using only the SIA module to segment the target regions of the query image results in a 0.19% increase in mIoU compared with the baseline. However, this performance decreases by 1.82% compared with the full PGSA module. This suggests that without utilizing discriminative prototypes to optimize the feature matching between the support and query images, the model becomes vulnerable to interference from complex background regions, leading to suboptimal performance. Using only the PC module results in a 1.38% increase in mIoU compared with the baseline, but this is 0.63% lower than the full PGSA module. This is because, during the feature extraction stage, some support foreground regions

TABLE VIII

EFFECTS OF THE SIA AND PC MODULES. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREE FOLDS COMPARED WITH THE BASELINE

| SIA | PC | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|---|
| ✓ | | 38.75 | 24.16 | 34.10 | 32.34 | 60.25 | +0.19 |
| | ✓ | 39.56 | 25.73 | 35.30 | 33.53 | 62.56 | +1.38 |
| ✓ | ✓ | **40.19** | **25.81** | **36.49** | **34.16** | **63.20** | +2.01 |

are inevitably lost due to operations such as convolution and pooling. If this lost information is not aggregated prior to the matching of support and query features, it will lead to incomplete segmentation of the target region. When both the SIA and PC modules are combined, the full potential of the PGSA module is realized, resulting in the model achieving its highest mIoU performance, which reaches 34.16%.

*3) Effects of TVCE Module:* This section focuses on the effectiveness of the TVCE module. As shown in Table VII, compared with the baseline, the mIoU and FB-IoU are improved by 2.42% and 1.95%, respectively. We also present qualitative visualizations of the activation maps generated by the TVCE module in Fig. 4(e). Specifically, for smaller objects such as "small vehicle" and "plane," the module accurately activates the target regions, while effectively distinguishing edge and detail features. Even for denser objects, such as "ship," the TVCE module achieves more accurate localization by leveraging class-aware alignment between textual and visual modalities. In contrast, the baseline and PGSA modules fail to prominently activate the target regions. The qualitative and quantitative results demonstrate that our module, without relying on limited support samples, can effectively extract semantic information from the query image by tightly aligning textual and visual features, thereby enhancing target region features and alleviating the issue of insufficient guidance when significant differences exist between the support and query images.

*4) Effects of Loss Function Weights $\alpha$:* As shown in Fig. 5, we evaluated the model's mIoU with different values of $\alpha$ in 0, 0.2, 0.4, 0.6, 0.8, and 1.0 under the one-shot setting across various folds. The model achieves the highest mIoU when $\alpha$ is 0.2. Therefore, to achieve the best performance for the model, we ultimately set the loss function weight to 0.2.

*5) Effects of OT:* To validate the superiority of incorporating an OT plan for aligning text and visual features, we compare it with the model before integration. As shown in Table IX, our method achieves a significant performance improvement, increasing mean mIoU by 1.91%. Specifically, before

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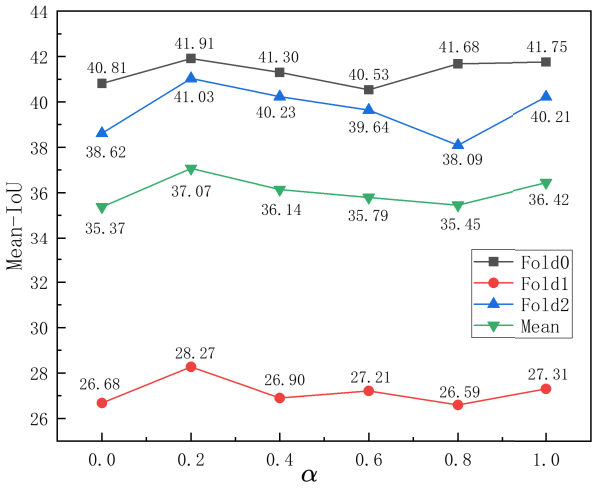



Fig. 5. mIoUs under different loss coefficients $\alpha$.

TABLE IX

RESULTS UNDER THE TVCE MODULE WITH OR WITHOUT THE INTEGRATION OF OT PLAN. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREE FOLDS COMPARED WITH THE TVCE MODULE WITHOUT THE INTEGRATION OF THE OT PLAN

| Method | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| TVCE (without OT) | 39.35 | 24.90 | 33.74 | 32.66 | 63.83 | - |
| TVCE (with OT) | **40.73** | **26.94** | **36.05** | **34.57** | **64.19** | +1.91 |
| TVCE (with OT [65]) | 40.28 | 26.62 | 35.34 | 34.08 | 63.75 | +1.42 |

integrating the OT plan into the TVCE module, the degradation in model performance stems from the fact that the CLIP model focuses primarily on high-level associations between images and text, rather than fine-grained visual features. Consequently, directly aligning textual and visual features via pretrained CLIP knowledge introduces inherent limitations. We further argue that this issue is exacerbated by the high interclass similarity inherent in remote sensing images. Therefore, we model the alignment of features from text and image modalities as the OT plan, which uses fine-grained visual features to update the text prompts, thereby injecting rich visual attributes into the text features. This minimizes the transport cost between the two modalities, achieving fine-grained semantic alignment.

Furthermore, we present segmentation results in Fig. 6 to validate the effectiveness of the OT plan in aligning prompts with visual features. It can be observed that the TVCE module without the OT plan only captures the coarse location of the target in the query image, resulting in blurry boundaries or incomplete segmentation. In contrast, integrating the OT plan enables the target prompt to focus on the most relevant pixel regions, thereby facilitating fine-grained alignment between textual prompts and visual features, which in turn leads to more accurate segmentation.

*6) Effects of Different OT Strategies:* In recent work, Bi et al. [65] introduced an OT-plan-based strategy that distributes semantic cues from multiple local masks to distinct agent prototypes, achieving promising performance. Therefore, to further validate the superiority of our cross-modal OT plan, we integrate their OT strategy into our TVCE module and conduct comparative experiments. As shown in Table IX, our method achieves the highest mean mIoU and FB-IoU. This

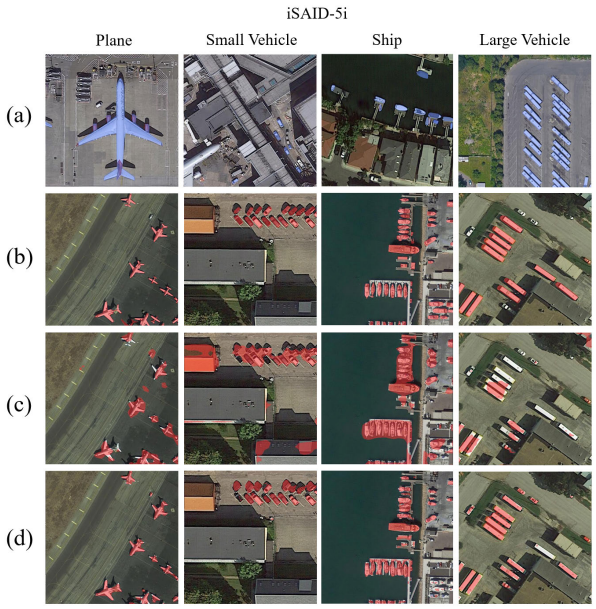



Fig. 6. Visualization of results with the OT plan on the iSAID-5i dataset. (a) Support image with the ground truth (blue). (b) Query image with the ground truth (red). (c) Results of our TVCE module (without OT) (red). (d) Results of our TVCE module (with OT) (red).

TABLE X

RESULTS UNDER DIFFERENT BACKGROUND SUBTRACTION METHODS. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREE FOLDS COMPARED WITH THE PGSA MODULE WITHOUT THE INTEGRATION OF THE SUBTRACTION

| Method | Fold-0 | Fold-1 | Fold-2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| without subtraction | 39.48 | 24.16 | 35.81 | 33.15 | 61.05 | - |
| with subtraction | **40.19** | **25.81** | **36.49** | **34.16** | **63.20** | +1.01 |

improvement primarily stems from our log-domain implementation of the Sinkhorn algorithm, which employs numerically stable log–sum–exp updates to avoid underflow and overflow when computing the transport kernel. As a result, the transport mass is more tightly concentrated on the highest similarity text–pixel pairs, yielding crisper, higher fidelity transport maps than the floating-point solver in [65].

*7) Effects of Subtraction Operation:* We explore the influence of the subtraction operation in the PGSA module on the accuracy of the model. As shown in Table X, incorporating the subtraction operation results in a greater performance improvement, increasing mean mIoU by 1.01%. This demonstrates that employing subtraction operations to eliminate ambiguous query regions, which exhibit high similarity to both support foreground and background features, facilitates the extraction of the most discriminative query foreground prototype. This prototype focuses on activating the query foreground regions most similar to the support foreground while reducing interference from similar backgrounds. Furthermore, a notable increase of 2.15% in the FB-IoU indicates that our method significantly improves the clarity between the foreground and background, thereby enabling more effective segmentation of the target region.

*8) Effects of Different CLIP Pretrained Models:* To evaluate the impact of different CLIP pretrained models on our method, we conduct experiments comparing it with RemoteCLIP [80].

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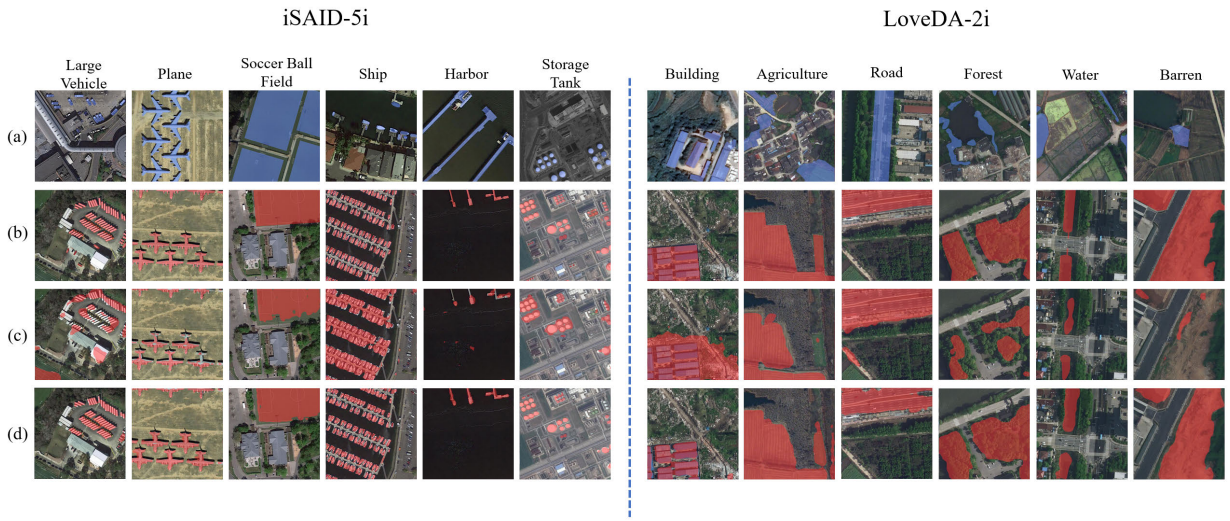



Fig. 7. Comparison of the model's segmentation performance on the iSAID-5i and LoveDA-2i datasets under the one-shot setting. (a) Support image with the ground truth (blue). (b) Query image with the ground truth (red). (c) Results of the baseline (red). (d) Results of our MVLPNet method (red).

TABLE XI

RESULTS UNDER DIFFERENT CLIP PRETRAINED MODELS. L-PARAMS AND PARAMS DENOTE THE NUMBER OF LEARNABLE PARAMETERS AND TOTAL PARAMETERS. INFERENCE SPEED DENOTES THE NUMBER OF IMAGES THE MODEL CAN INFER PER SECOND

| Method | 1-shot | 5-shot | L-Params | Params | Inference Speed |
|---|---|---|---|---|---|
| MVLPNet (CLIP-ViT-B/16) | 37.07 | 41.46 | **8.9M** | **205.1M** | 7.4 |
| MVLPNet (CLIP-ViT-B/32) | 36.92 | 41.57 | 8.9M | 206.8M | **8.3** |
| MVLPNet (RemoteCLIP-ViT-B/32) | **38.04** | **41.89** | 8.9M | 206.8M | 7.6 |

The results are shown in Table XI. We observe that the CLIP model (ViT-B/32) fine-tuned on remote sensing images outperforms the CLIP model (ViT-B/32) pretrained on natural images, achieving an improvement in mean mIoU of 1.12% and 0.32% for one-shot and five-shot settings, respectively. However, this improvement also introduces additional computational cost. Compared with the original CLIP model (ViT-B/32) pretrained on natural images, RemoteCLIP exhibits a decrease of 0.7 frames/s in inference speed. This is primarily due to the overhead introduced by domain-specific pretraining on large-scale remote sensing image–text pairs, as well as the use of FP32 for inference (in contrast to the original CLIP's FP16 implementation), both of which increase computational complexity during cross-modal inference.

*9) Visualization:* Fig. 7 presents the model's prediction results under the one-shot setting on the iSAID-5i and LoveDA-2i datasets. Our method produces more accurate segmentation results compared with the baseline. Specifically, in the iSAID-5i dataset, as observed in the first column, the appearance of large vehicles is similar to that of background buildings. Compared with the baseline, which incorrectly activates the background building regions, our method effectively distinguishes between the features of the two, achieving more accurate segmentation. In addition, in the fourth and sixth columns, the baseline method frequently merges the ship and storage tank predictions, leading to unclear boundaries, while our model effectively distinguishes these boundaries. For the more challenging LoveDA-2i dataset, the baseline method erroneously activates regions prone to confusion, such as building and road, resulting in incomplete masks.

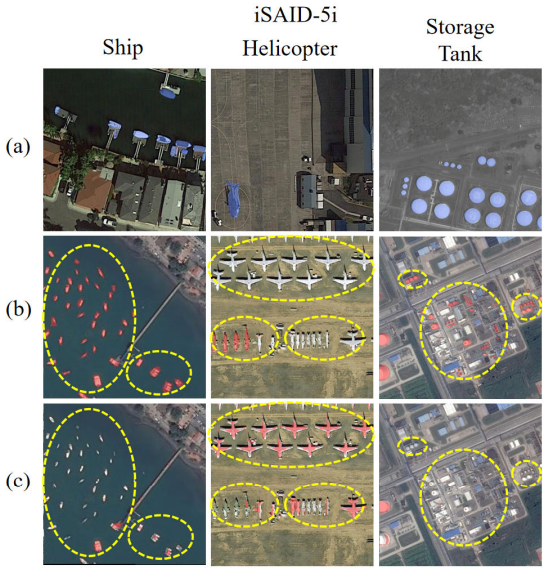



Fig. 8. Visualization of segmentation failure results under the one-shot setting on the iSAID-5i dataset. (a) Support image with the ground truth (blue). (b) Query image with the ground truth (red). (c) Results of our MVLPNet method (red).

In contrast, our approach captures representative class information to optimize feature matching, thereby generating more accurate predictions.

*10) Failure Case Analysis:* Although the proposed MVLPNet achieves state-of-the-art segmentation performance on the iSAID-5i and LoveDA-2i remote sensing datasets, it still exhibits several failure segmentation cases. As shown in Fig. 8, when the target is extremely small and does not stand out against the background, our method fails to segment it (e.g., the ship and storage tank highlighted by yellow dashed lines). In addition, when both planes and helicopters appear in the same image, the proposed method fails to distinguish between them because of the high similarity in their textual semantics and visual appearances, resulting in incorrect segmentation. To address these problems, we propose some thoughts.

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1) Employing diffusion models [81], [82] to generate additional high-quality remote sensing images based on support samples to enhance the model's generalizability.
2) Representative and learnable textual prompts are introduced to help the model better focus on the target region.

## V. Conclusion

In this article, we propose an MVLPNet that employs a collaborative optimization strategy for visual–textual features to tackle the remote sensing FSS task. Specifically, to overcome the limited support set's inability to capture intraclass diversity, we first propose a TVCE module that achieves tight alignment between the category-specific text embeddings and the visual features of the query image based on the OT plan, thus mining category semantics from the query image itself for better guiding the query segmentation. Furthermore, to alleviate interference caused by complex background regions, the PGSA module explicitly guides the model to focus more on the consistent foreground regions between the query and support images rather than the background ones. Finally, our method achieves state-of-the-art performance on the iSAID-5i and LoveDA-2i datasets. In future work, we will further explore the potential and feasibility of multimodal features in the FSS domain.

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